ktwu@utexas.edu
ktwu01 ktwu01

Koutian Wu

(+1) 737-420-8027
Flexible with relocation

2+ years work experience (U.S./International); **4+** years research/programming experience; **302** stars on GitHub.

Education

Ph.D. Student in AI for Geosciences | **University of Texas at Austin** | 2024 – Expected: May 2029

Reviewer for *ACM CSUR*, *ACM AgentSkills*, *COLM Agent Behavior*, *ACM TOSN*, *IEEE TNNLS*, and *PSS*

B.S. in Geosciences (Honors Class) | **University of Science and Technology of China** | 2020 – 2024

Finalist for the Guo Moruo Scholarship (USTC's highest undergraduate honor)
Founded one of the largest student clubs, grew from 0 to 1,300+ members. [News]

Work Experience

Lead Developer, ESM-bench for Earth System Models — UT-Austin | Dec. 2025 – Present

- Designed a 243-task benchmark across 3 Earth System Models (Noah-MP, CLM5, MOM6) evaluating whether LLM agents can bridge the physics-and-code understanding gap in 200k+ line Fortran/C codebases.
- Built a multi-model evaluation framework testing 5+ LLMs (GPT-5.5, Claude Opus, Gemini, etc) with 3 progressive hint levels and a 6-item physics-aware semantic and result-based rubric.
- Achieved <20% exact match across all frontier models, demonstrating that physical correctness in scientific code remains an unsolved challenge for current LLMs. [Preprint]

Co-Lead, FrontierShiftBench — ScienceIntelligence.org | Jun. 2026 – Present

- Co-lead an AI-science benchmark that evaluates research agents on both dry computational tasks and wet-lab planning tasks, covering the full spectrum from simulation/analysis to experimental protocol design.
- Owned the task interface: drafted the task schema, contribution guide, public/hidden task split, validators, and scoring rubrics that external contributors must follow.
- Built the expert-review workflow for auditing agent-generated tasks before the format freeze, defining what evidence a candidate task must show to be accepted.

Program Committee — Proposed NeurIPS 2026 Workshop | May 2026 – Present

- Support proposal and program review for The Future of the AI Research Ecosystem, focused on benchmarks, datasets, evaluation protocols, artifact standards, and AI-scientist methods.
- Shape workshop topics on publication formats, citation graphs, review workflows, archives, and agent-verifiable research infrastructure.

Core Contributor, ASI-Bench — Apex Intelligence | Apr. 2026 – Jul. 2026

- Contributed to a project-level AI-for-science agent benchmark where ground-truth generation, task isolation, and scorer fidelity determine whether agent evaluations reflect real scientific capability. [Paper] [Website] [Code]
- Landed 9 merged PRs, including 3 calibrated earth-science tasks with parametric ground-truth generators, B1–B4 prompt tiers, task sandbox runtimes, and custom process scorers.
- Hardened evaluation validity by fixing scorer CSV/correlation semantics, removing prompt/scorer leakage, relieving hidden-criteria penalties, and tightening hydrology scoring against generic soil-bucket solutions.
- Listed as a contributor on the ASI-Bench paper (60 project-level tasks across 11 scientific domains with B1–B4 guidance-withdrawal evaluation).

Task Contributor, ResearchClawBench Earth/ESM Task — ResearchClawBench | Jun. 2026

- Contributed `Earth_004`, a Noah-MP hydrology task on surface runoff and runoff/infiltration partitioning.
- Built the task package with context: an 81-case diagnostic CSV, a 5-item scoring checklist, and 2 target figures.

Open-Source Contributor, IKP Evaluation Suite — 19PINE-AI/ikp #4 | Jul. 2026

- Landed a 7k+ line IKP suite adding a pre-run cost estimator, adversarial-robustness analysis, gaming-resistant v2 estimator, validation harness, figures, and research-proposal docs.
- Designed held-out probe-split estimation to reduce public-ladder overfitting; validation reproduced recorded accuracies for 201 models and reported R^2=0.91 calibration quality.
- Unified divergent calibration constants so v1/v2 estimators and analysis scripts load the same fitted artifact, removing up to 3x frontier-scale inconsistency.

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Open-Source Contributor, AI Safety Audit Infrastructure meridianlabs-ai/inspect_petri | Jun. 2026 – Present

- Contributed to Petri, an alignment-auditing agent for evaluating misaligned model behavior, where transcript fidelity and benchmark integrity determine whether safety findings are credible.
- Landed 2 tested upstream fixes strengthening audit evidence integrity: judge-visible auditor prefill attribution and shipped benchmark-seed canary coverage.
- Opened additional tested PRs to surface fail-open realism approvals and judge non-score coverage, plus a provenance-manifest proposal linking findings to model, prompts, run config, and version.

Open-Source Contributor, ARA Agent-Native Research Skills ARA-Labs/ARA | Jun. 2026

- Contributed 2 upstream commits to `@ara-commons/ara-skills`: fixed canonical npm metadata [Commit] and repaired stale `.ara-skills.json` `updatedAt` writes with a regression test [Commit].

Co-Lead & Maintainer, ResearchSkills.ai ScienceIntelligence.org | Remote | Mar. 2026 – Present

- Architected and built the production web server (Node.js + Express, React + Vite, Supabase Postgres, GitHub OAuth) powering the submission-to-publication pipeline, contributing 207 of 250 commits (about 83% of the codebase) as top contributor. [Site] [Org]
- Designed the contributor-draft/admin-approve workflow so every accepted skill reaches the public ScienceIntelligence/ResearchSkills registry through authenticated review; sustained 200+ atomic, end-to-end-verified commits in about four weeks.

AI Intern, AI Agent Knowledge Base Evaluation 19Pine.AI (Singapore) | Remote | Jul. 2025 – Aug. 2025

- Implemented a multi-dimensional evaluation system from scratch for PineAgent's RAG knowledge base.
- Developed a data sanitization module removing 2,000+ PII entries and noise from 1,000+ call sessions.
- Extracted Q&A pairs (knowledge/method/strategy) from call sessions to form the evaluation dataset.
- Deployed a concurrent LLM-as-judge + rule-based engine, measuring P/R/F1 to enable RLHF optimization.

Research Assistant, ML Model/Multi-agent Development UT-Austin | Austin, TX | Aug. 2025 – Jan. 2026
Project Lead, High-Performance Computing Allocation NSF NCAR | Remote | Aug. 2025 – Present

- Proposed and secured NSF NCAR's 1k GPU + 22k CPU hours high-performance computing allocation.
- Integrated machine learning parameter calibration for a SOTA physics-based land surface model (Noah-MP).
- Building a multi-expert AI agent system for automated parameterization for physics-based climate models.

Co-Lead, AI-Agent Skills for Earth-System Models earth-space-ai | May 2026 – Present

- Built AI-agent onboarding skills for 10 models: E3SM, WRF, CAM, MOM6, Noah-MP, and others.
- Proposed upstream docs pointers to expose AI-assisted onboarding without changing model code.
- Designed a Texas 12.5 km, NLDAS-2, and TACC run plan for Noah-MP undergraduate onboarding.

Open-Source Contributor, Git git/git | Jun. 2026

- Authored and landed a patch in git/git through the Git mailing list; fixed an invalid `.gitattributes` line-ending attribute (`eof=lf` → `eol=lf`) so Perl scripts normalize to LF on checkout. [Commit]

Publications

- Liu, X. et al. (including **Wu, K.**). From Personas to Simulated Users: A Fitness-for-Purpose Survey. (Submitted to AAAI 2027 Artificial Intelligence for Social Impact Track)
- Li, X. et al. (including **Wu, K.**) (2026). MatrAIx: Simulating the World with 8.3 Billion Persona Agents. (Preprint) [Paper] [Demo] [Code]
- Xu, W. et al. (including **Wu, K.**) (2026). ResearchClawBench: A Benchmark for End-to-End Autonomous Scientific Research. (Submitted to AAAI 2027) [Paper]
- Zhou, J. et al. (including **Wu, K.**). ASI-Bench: At the Dawn of Artificial Superintelligence. (Preprint) [Paper] [Website] [Code]
- Mai, G., Lao, N., Zhang, J., Mao, L., Wang, Z., Wu, N., Janowicz, K., **Wu, K.**, Rao, J., Gao, S., & Zhu, R. (2026). On the Ethics of Generative GeoAI: Explainability, Bias, Hallucination, Accountability, Privacy, and Trust. In *Geography According to Foundation Models*, Vol. 422, pp. 215–232. IOS Press. [Chapter]
- **Wu, K.**, Cao, Y., & Mai, G. (2026). ESM-bench: A Benchmark for Evaluating Whether AI Agents Understand Earth System Model Physics and Code. (In prep to NeurIPS dataset). [Preprint]

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- **Wu, K.**. Noah-Agent: A Multi-Expert AI Agent Framework for Automated Parameterization and Validation of Large-Scale Fortran Climate Models. (In prep). [Record]
- **Wu, K.**, Yi, W.*, Xue, X.*, Reid, I., & Lu, M. (2024). Diurnal and Seasonal Variations of Meteor Speed and Arrival Angle Observed by Mengcheng Meteor Radar. *JGR: Space Physics*. [Paper] [Code] [Data]
- **Wu, K.***, Xu, X., Jiang, J., & Shen, A. (2024). A Summary Report on the Space Physics Practical Education in 2022. *Reviews of Geophysics and Planetary Physics*. [Paper] [Code] [News]

Skills

- Python, FastAPI, JavaScript, TypeScript, React, SQL, PyTorch, LangChain, RAG
- Claude Code, Codex, Openclaw, Hermes, Cursor, Docker, AWS/GCP/Azure, Redis, Git, HuggingFace
- Multi-model LLM/VLM evaluation, benchmark & dataset construction, LLM-as-judge rubric scoring, OpenAI/Anthropic/AWS Bedrock APIs, error taxonomy & failure analysis